Body Control Module - High-Level Design (Rev A)

1. Component Overview

This document describes the High-Level Design of the Body Control Module software cluster, covering lighting, door, and w

Software Components

Component: BodyControlModuleComponent
Component: LightingManagerComponent
Component: DoorControlComponent
The BodyControlModuleComponent acts as the central coordinator for all body domain functions.

2. Interfaces and Ports

Interface: LightingControlInterface
Interface: DoorStatusInterface
PPort: LightingManager_PPort
RPort: DoorControl_RPort
LightingManagerComponent provides LightingControlInterface via LightingManager_PPort. DoorControlComponent requir

3. Signals

Signal: HeadlampState_Signal
Signal: DoorAjar_Signal
Signal: IndicatorRequest_Signal

4. Dependencies

BodyControlModuleComponent depends on LightingManagerComponent for lighting state and on DoorControlComponent

5. Functional Flow

On receiving an IndicatorRequest_Signal, LightingManagerComponent evaluates the current HeadlampState_Signal and i